

DiaTool-DPO: Multi-Turn DPO for Tool-Augmented LLMs

Enhancing Tool-Augmented LLM Dialogue via Direct Preference Optimization

Automatic paired-trajectory construction • 5-state MDP • Near-GPT-4o performance on open-source models

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Tool-Augmented LLMs must control conversation flow, but SFT-only training exhibits critical failure modes:

- Hallucinated tool calls — The model invokes tools with missing or fabricated arguments instead of asking follow-up questions.
- Failed slot-filling — Incomplete user queries are not recognized; the model skips multi-turn QA needed to gather required parameters.
- Missing rejection — Out-of-scope or unsupported requests are not rejected, leading to incorrect tool invocations.
- SFT limitation — Expert-trajectory supervision alone does not teach the model to distinguish correct vs. incorrect dialogue flows.

Core Insight

Direct Preference Optimization (DPO) can teach dialogue control by contrasting correct trajectories against incorrect ones. DiaTool-DPO automatically constructs these paired trajectories without human labeling, targeting the exact failure modes SFT misses.

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process

Internal (unobserved) states govern when to ask, call, wait, or complete:

1. Initial

No history. Out-of-scope requests return rejection and remain here.

2. Tool Selected w/o Complete Slots

Tool identified but required parameters missing. Slot-filling QA occurs here.

3. Tool Selected w/ Complete Slots

Tool selection and all argument values are determined. Ready to call.

4. Wait for Tool Response

Tool call executed; system awaits results.

5. Complete

Tool results received; completion message generated for the user.

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Example
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	User provides all required arguments upfront. No slot-filling needed.
Type 2	$1 \rightarrow (2 \times N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	Query lacks some arguments. Slot-filling QA repeats N times until all fields are gathered.
Type 3	$1 \rightarrow 1$	Requested functionality is unavailable. Tool call must be rejected immediately.

Automatic Paired Trajectory Construction Without Human Labels

Chosen (correct) vs. Rejected (incorrect) trajectories teach specific dialogue-control lessons:

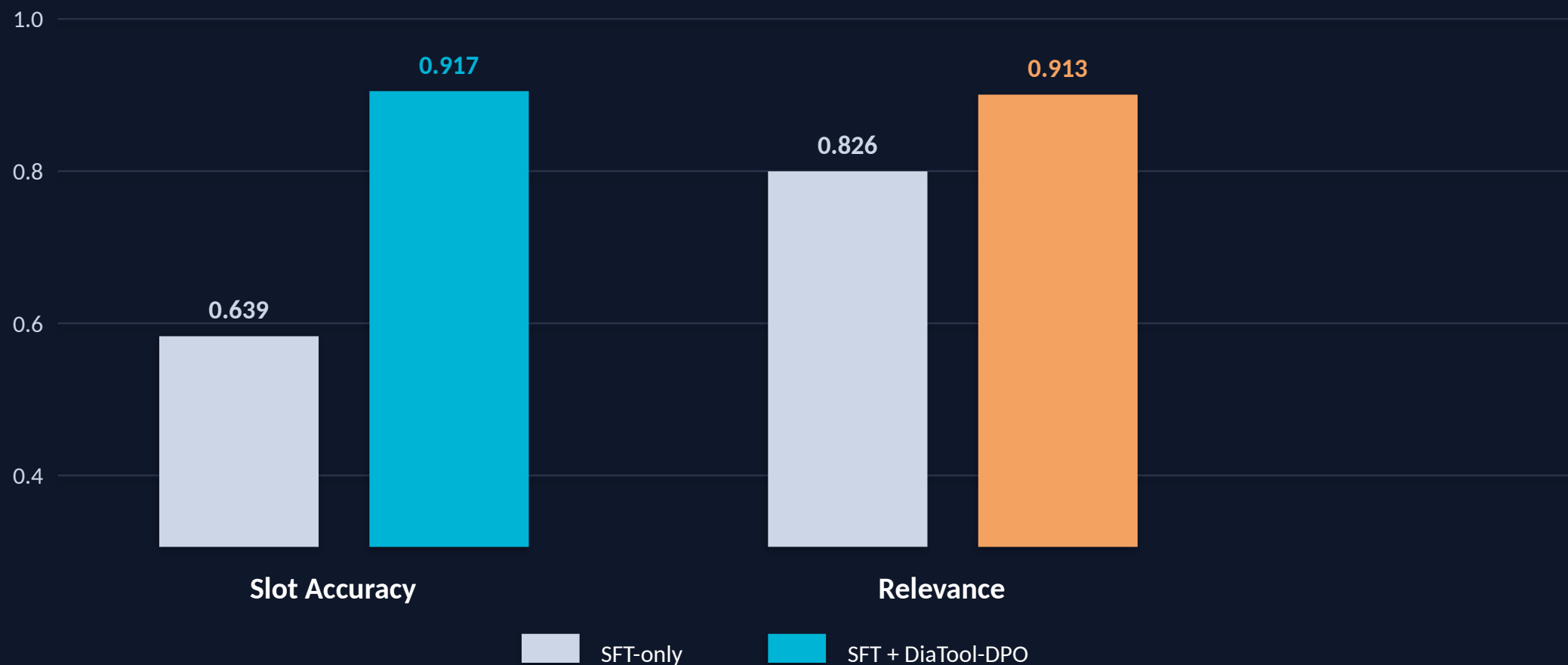
Query	Chosen	Rejected	Learning Lesson	Count / Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089 / Easy
Type 1	1→3→4→5	1→(2*N)→3→4→5	Prevent redundant slot-filling	562 / Hard
Type 1	1→3→4→5	1→(2*M)→3→4→5	Prevent redundant slot-filling	2,530 / Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,090 / Easy, 562 / Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089 / Easy
Type 2	1→(2*N)→3→4→5	1→3→4→5	Prevent slot hallucination	562 / Hard
Type 2	1→(2*N)→3→4→5	1→(2*M)→3→4→5	Prevent slot hallucination	2,530 / Hard
Type 2	—	1→1	Tool call accept	2,089 / Easy, 562 / Hard
Type 3	1→1	1→3→4	Tool call reject	567 / Hard
Type 3	1→1	1→(2*N)→3→4	Tool call reject	562 / Hard

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO

Adding DiaTool-DPO to SFT yields the largest absolute gains on slot-filling and tool-rejection accuracy:



Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

1. MDP formulation enables systematic preference pair construction — Defining 5 internal states and 3 query types allows automatic generation of chosen/rejected trajectory pairs without human annotation.
2. Automatic dataset construction eliminates labeling cost — Every pair teaches a targeted lesson (redundant slot-filling, slot hallucination, tool call reject/accept) and scales to new domains.
3. DPO complements SFT; it does not replace it — DiaTool-DPO-only performs poorly in isolation. The combined SFT + DPO pipeline achieves the best balanced (Macro Average) results.
4. Largest gains are in slot-filling and tool rejection — LLaMA-3-8B improves Slot accuracy by +43.5% and Relevance by +10.5%, reaching near-GPT-4o performance (94.8% info gathering, 91% rejection).
5. Language-agnostic and production-ready — Primary experiments were conducted in Korean, yet the method generalizes to English, opening broad applicability for production TA-LLMs.